Activity - 3

1) Gradient Magnitude

$$G = \sqrt{Gx^2 + Gy^2}$$

$$= \sqrt{24^2 + 18^2}$$

$$= \sqrt{576 + 324}$$

$$= \sqrt{900}$$

$$= 30$$

Gradient direction (θ).

$$\theta = \arctan\left(\frac{Gy}{Gx}\right) = \arctan\left(\frac{18}{24}\right)$$

$$= \arctan(0.75) \approx 36.87^\circ.$$

2) Non-Maximum Suppression.

Given magnitudes:

Pixel	P_1	P_2	P_3	P_4	P_5
Magnitude	25	50	90	60	30

→ Since the gradient is horizontal, compare each pixel with its left & right neighbours.

→ P_1 → suppressed (border).

→ P_2 → $50 < 90$ → suppressed.

→ P_3 → $90 > 50$ → Retained.

→ P_4 → $60 < 90$ → suppressed.

→ P_5 → suppressed.

→ Only P_3 (90) remains after Non - maximum suppression.

3) Double Threshold classification.

High threshold = 100
Low threshold = 50.

Pixel	Magnitude	classification
P_1	135	strong
P_2	90	Weak
P_3	45	Non-Edge
P_4	120	strong
P_5	70	Weak
P_6	20	Non-Edge

b) Hysteresis thresholding:

Magnitude:

Pixel	Magnitude	Final classification
P1	130	Strong
P2	95	Weak
P3	40	Non-edge
P4	115	Strong
P5	65	Weak
P6	25	Non-Edge
P7	105	Strong

→ P2 is connected to P1 → Keep as an edge.